

TOI-1811b

R-2190

Benchmark run · workflow W8-benchmark-deep · record deep-bench_TOI1811_250413 · created 2026-08-13T23:46:57+00:00

BENCHMARK: ACCURATE — fit consistent with published values

Results

Mid-Transit Time (Tmid)	2460778.667 +/- 0.018 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.162 +/- 0.047
Transit depth (Rp/Rs)^2	2.62 +/- 1.51 [%]
Orbital Inclination (inc)	85.81 +/- 2.39
Airmass coefficient 1 (a1)	1.134 +/- 0.035
Airmass coefficient 2 (a2)	-0.103 +/- 0.058
Scatter in the residuals of the lightcurve fit is	3.05 %
Best Comparison Star	#2 - [619, 253]
Optimal Aperture	3.61
Optimal Annulus	12.99
Transit Duration (day)	0.043 +/- 0.033

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.162 ± 0.047 vs 0.13272 (deviation 0.62σ)
Duration fit vs published	0.043 d vs 0.0813 d (deviation 1.16σ)
Mid-time O–C	-23.3 min
Depth SNR	3.4
Residual scatter	3.05 %

Light curve

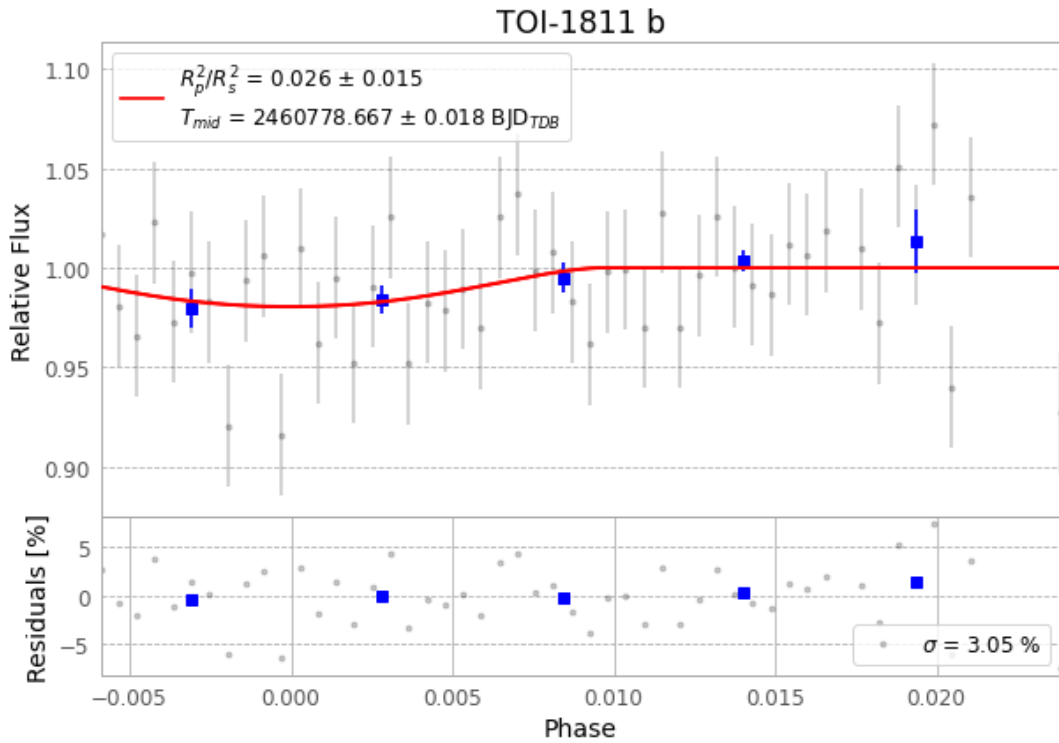


Figure 1 — FinalLightCurve_TOI-1811 b_2025-04-12.png (SHA-256 5f177c962fcdcb20...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [342, 197], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 d9a74f7f22331d95...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit 1dc3d22

Data provenance

57 FITS frames (37 MB), TOI1811250413032116.FITS → TOI1811250413060915.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-TOI1811-250413.log (31a88567fd02...), FinalLightCurve_TOI-1811 b_2025-04-12.png (5f177c962fcd...), FinalLightCurve_TOI-1811 b_2025-04-12.csv (df7e482e3a59...)

Compute resources

Wall time 287.7 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-13 23:46Z.